**FUNCTIONS QUIZ**

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Name: Solutions

**1.** Find the value of $b$ that makes the function

$$f(x) = \begin{cases} \dfrac{x^2-4}{x-2} & \text{if } 2 < x \\ 2x+b & \text{if } x \leq 2 \end{cases}$$

a continuous function.

Since $\frac{x^2-4}{x-2}$ is continuous on $(2,\infty)$ and $2x+b$ is continuous on $(-\infty,2]$, we just need to ensure that $f$ is continuous at 2; that is

$$\lim_{x\to 2} f(x) = f(2) = 2(2)+b = 4+b.$$

$$\lim_{x\to 2^+} f(x) = \lim_{x\to 2^+} \frac{x^2-4}{x-2} = \lim_{x\to 2^+} \frac{(x-2)(x+2)}{x-2} = \lim_{x\to 2^+} x+2 = 2+2 = 4.$$

$$\lim_{x\to 2^-} f(x) = \lim_{x\to 2^-} 2x+b = 4+b$$

Since $\lim_{x\to 2} f(x)$ exists if and only if $\lim_{x\to 2^+} f(x) = \lim_{x\to 2^-} f(x)$, we need

$$4 = 4+b$$

Therefore $\boxed{b=0.}$